Balance each direction, not just the force sizes

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Equilibrium / ES: equilibrio / PT: equilíbrio

Zero a : forces sum to zero on each axis.

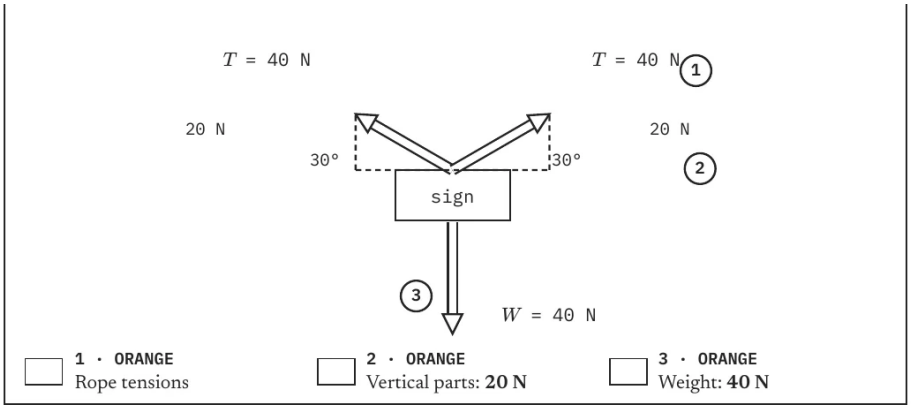
Also on your note page: Balance, Symmetry, Rest.

B READ THE RELATIONSHIP

Equilibrium requires zero force sum on every axis

As you read, find: balance · upward part · weight.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 40 N sign hangs at rest from two equal ropes, each 30° above horizontal. Find the tension in each rope.

Given W = 40 N ; θ = 30° equal tensions a = 0

Rule x parts cancel. 2 T sin 30° = 40 N .

Work 2 T (0.5) = 40 N T = 40 N

Answer 40 N per rope Each rope gives 20 N upward.

E SAVE FOR THE FINAL MEETING

Zero net force would give XJ a constant velocity, not a guaranteed stop. Analysts must check each coordinate direction before treating its motion as steady.

Read the physics: Equilibrium

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Choose horizontal and vertical axes. Add the forces on each axis. When both sums are zero, acceleration is zero. The object can be at rest or moving steadily.

02 HOW IT WORKS

A hanging sign has downward weight and upward support. If the ropes are vertical, add their tensions. If the ropes are tilted, only the upward parts support the weight.

03 WHAT TO CHECK

Being at rest for an instant is not enough to prove equilibrium. Look for zero net force and unchanged velocity.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 40 N sign hangs at rest from two equal ropes, each 30° above horizontal. Find the tension in each rope. 40 N per rope Each rope gives 20 N upward.

Model check: Equilibrium requires zero force sum on every axis.

05 CONNECT TO XJ

Zero net force would give XJ a constant velocity, not a guaranteed stop. Analysts must check each coordinate direction before treating its motion as steady.

SAY IT BEFORE YOU WRITE

Can an object in equilibrium be moving? The ___ forces balance. Its acceleration is ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Equilibrium

ES: equilibrio / PT: equilíbrio

Meaning in my words	How can you tell?
How would you explain equilibrium to a friend? In my words, this means ___.	A cart rolls straight at steady speed. What must the horizontal forces add up to? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left. My example shows ___.	A tossed ball is briefly still at the top. Is it in equilibrium there? I would change ___ because ___.

Balance

ES: equilibrio de fuerzas / PT: equilíbrio de forças

Meaning in my words	How can you tell?
How would you explain balance to a friend? In my words, this means ___.	Two cables suspend a sign at rest. What must their upward components add up to? I can tell because ___.
Example or sketch	**Non-example and why**
A sign weighs 12 N and two cables share the upward support equally. What is each vertical contribution? My example shows ___.	The horizontal components cancel. Is that enough to show equilibrium? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Symmetry

ES: simetría / PT: simetria

Meaning in my words	How can you tell?
How would you explain symmetry to a friend? In my words, this means ___.	Two identical cables hold a centered sign at equal angles. What tells you their tensions match? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch equal tension arrows. What happens to the two horizontal components? My example shows ___.	One cable is much steeper. Can you still assume the tensions are equal from symmetry alone? I would change ___ because ___.

Rest

ES: reposo / PT: repouso

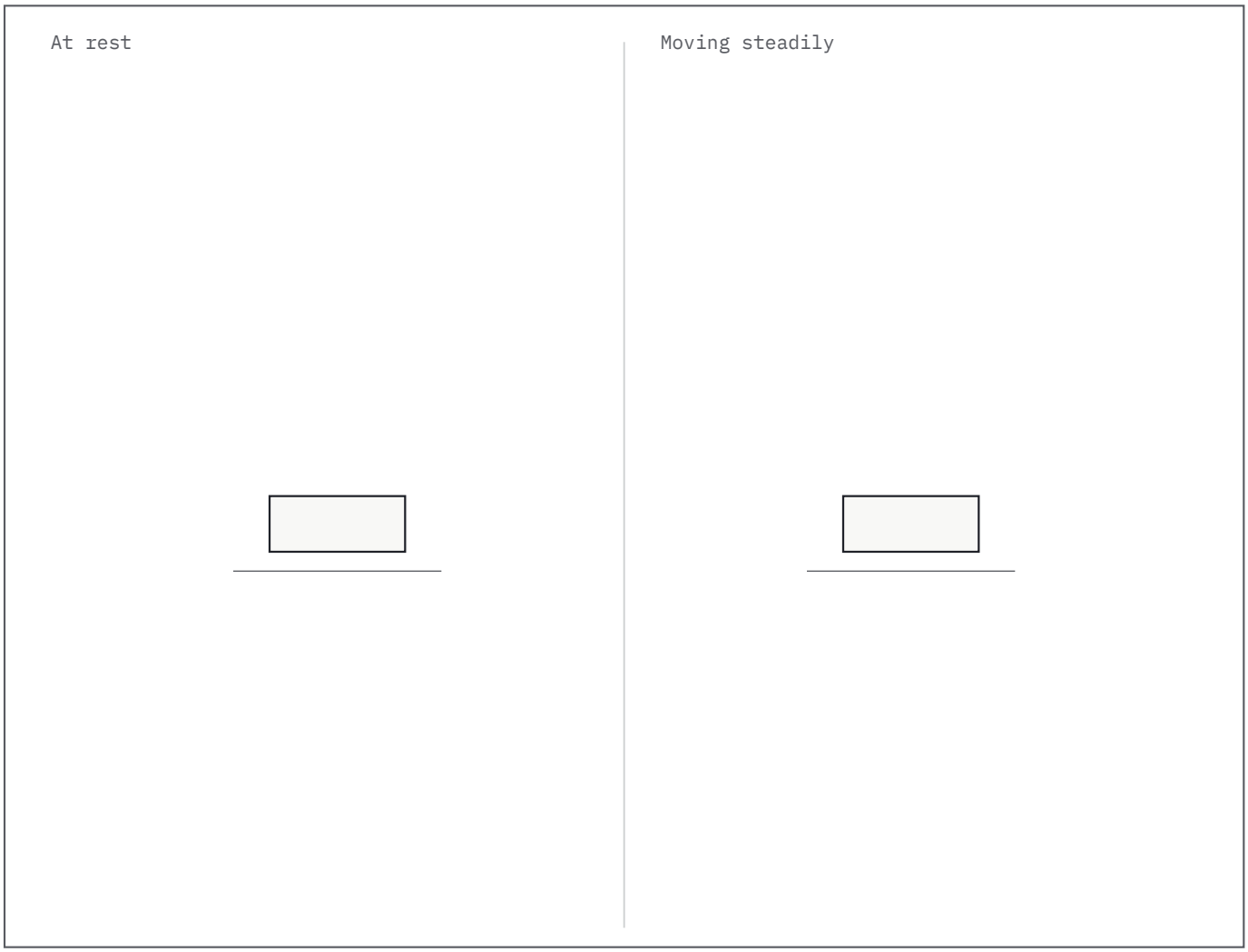
Meaning in my words	How can you tell?
What does it mean for an object to be at rest at an instant? In my words, this means ___.	At the top of a ball's straight upward toss, what is zero and what is still acting? I can tell because ___.
Example or sketch	**Non-example and why**
How would you draw velocity and acceleration at that instant? My example shows ___.	"It stopped, so the forces must balance." What does the next instant show? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch two systems with balanced forces: one at rest and one moving steadily. Label the common acceleration.

Labels you can use: balance · upward part · weight.



Tell a partner

The ___ forces balance. Its acceleration is ___.

Stopped or coasting?

18

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish zero acceleration from zero velocity.

CHOOSE ONE JOB TOGETHER

FLIGHT CONTROLLER ☐

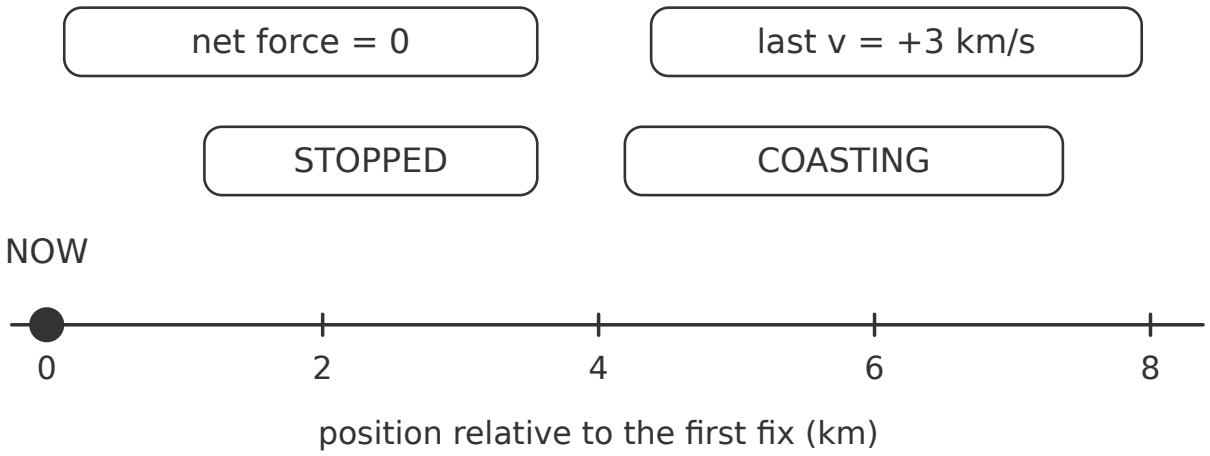
Choose the status graphic. Mark where XJ should be two seconds later relative to the first sighting.

BROADCAST PRODUCER ☐

Choose the on-air label. Explain why zero acceleration and zero speed do not mean the same thing.

Word help: balanced / coasting / acceleration

The flight display reads "net force zero." XJ's last velocity was **+3 km/s**. Two graphics say STOPPED and COASTING.



01 OUR FIRST CALL

Which status should go live? Use a second position to support your call.

The status is ___ because net force ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Can the old status describe the next interval? Explain what the new report changes.

Now we would say ___ because ___.

Switch speakers. In two dimensions, what must the force sums be along both axes for the forces to balance?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks